Name: _____ Date: _____ Period: _____

Unit 1 - Quiz #1 – EXTRA HELP

1) Find the value of n that makes this statement true:

$$\mathbf{2^3 \cdot 2^n = 2^7}$$

2) Find the value of n that makes this statement true:

$$\mathbf{\frac{2^9}{2^n} = 2^3}$$

3) What is the area of the rectangle below?



4) Students were asked to simplify the expression using laws of exponents:

$$\mathbf{5^4 \cdot 5}$$

Who was correct?

a) Olivia wrote 25^4 c) Chris wrote 5^5

b) Pete wrote 25^5 d) Dawn wrote 5^4

5) Students were asked to simplify the expression using laws of exponents:

$$\mathbf{\frac{8^{10}}{8^2}}$$

Who was correct?

a) Tim wrote 8^5 c) Hannah wrote 1^5

b) Bryce wrote 8^8 d) Jess wrote 1^8

6) Carla multiplied the following: $(2a^5)(4a) = 6a^6$

Is Carla correct? Explain how you determined your answer.

7) Lauren says that $(-2)(-2)(-2)(-2)(-2)(-2) = -2^6$. Is Lauren correct in her notation? Why or why not?

8) Simplify the following. Leave you answer in exponential notation.

$$\mathbf{4^6 \cdot \frac{4^4}{4^3}}$$

9) Given the expression below, what is an equivalent expression in exponential form?

$$\mathbf{-3 \cdot -3 \cdot -3 \cdot -3 \cdot b \cdot b}$$

a) -3^4b^2

b) $-3b^6$

c) $(-3b^2)^4$

d) $(-3)^4b^2$

10) Simplify the expression using the laws of exponents. Leave your answer in exponential form

$$\mathbf{\frac{(6^5 \cdot 6^6)}{6^3}}$$